Maximum likelihood and linear regression

Machine Learning, lecture 2

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YMLS Uzbekistan, fall 2026

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Two loose ends from last week

We used both without an explanation.

0.37, 0.65

The learned weights for the five students. Where did they come from?

MSE

Why do we square the errors and not count them some other way?

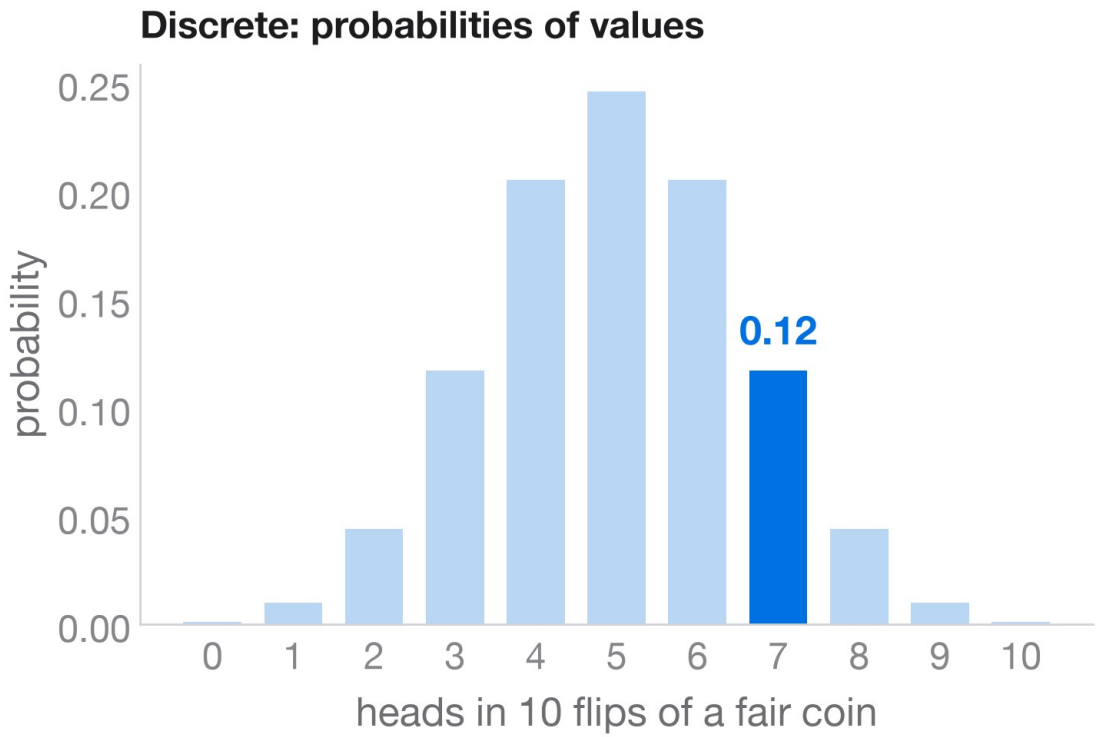
Today

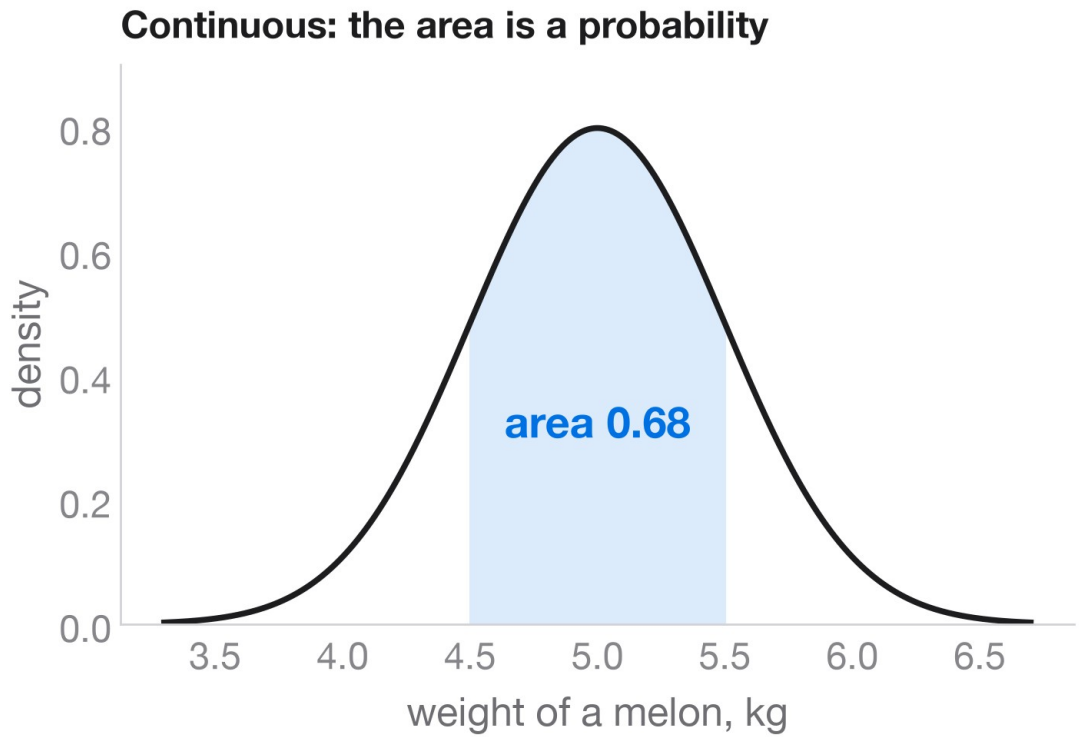
1. The probability we need
2. Maximum likelihood
3. Why squared error
4. Linear models and linear regression

The probability we need

A random variable has a distribution

It says how likely each value is.





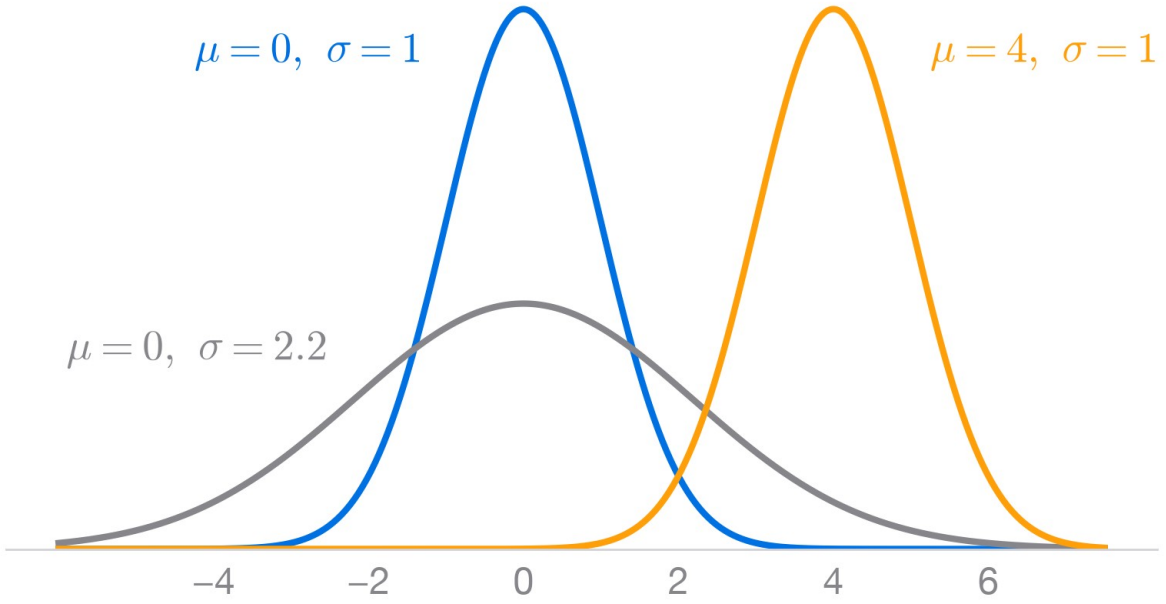
Expectation and variance

$$\mathbb{E}[X] = \sum_x x\, p(x)$$

The centre of the values.

$$\mathrm{Var}[X] = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \sigma^2$$

How far the values spread around the centre.

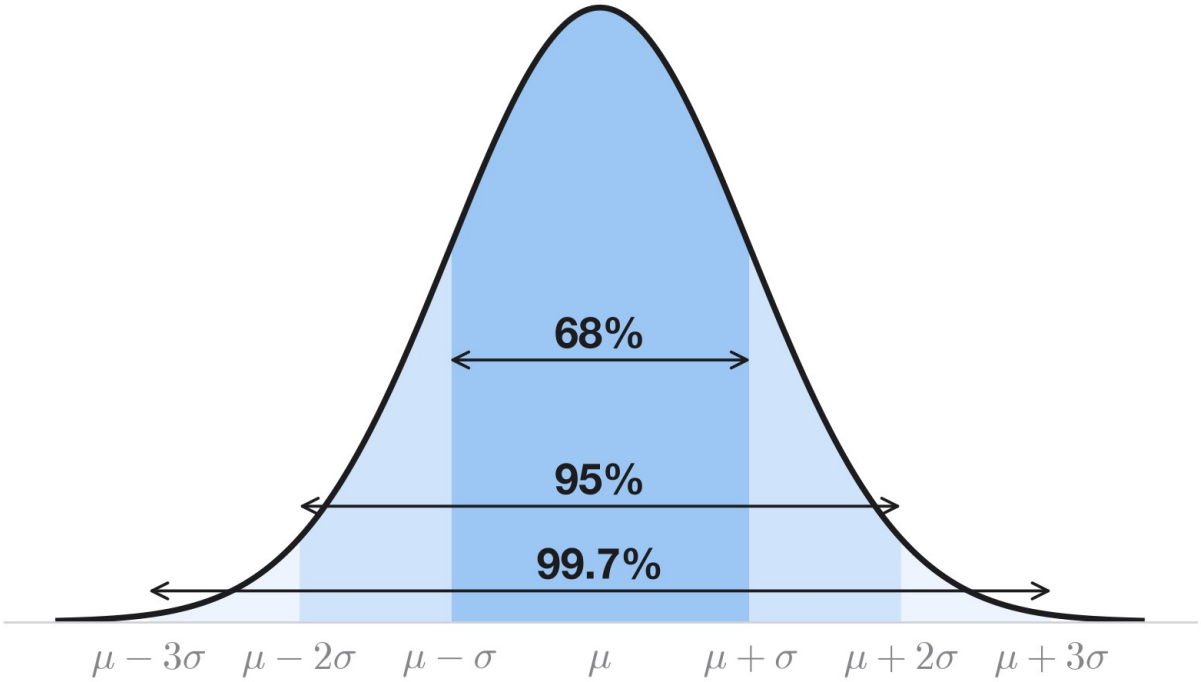


The normal distribution

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

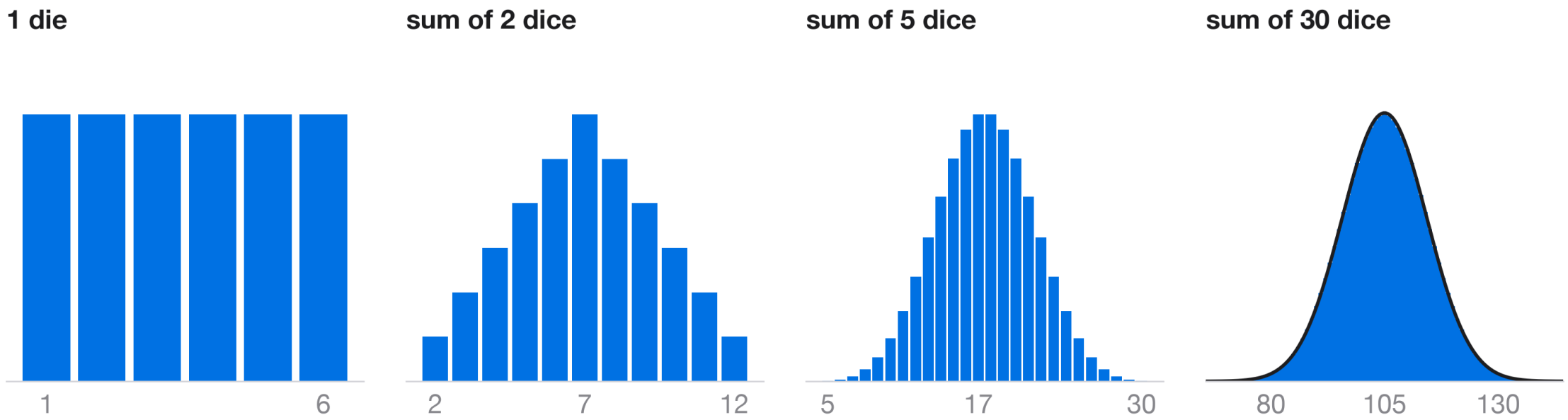
$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Two parameters: the centre μ and the spread σ.



Why the bell is everywhere

Add up many small independent effects and the sum is close to normal.



The central limit theorem. Measurement noise is such a sum.

Independent and identically distributed

i.i.d.: every object comes from the same distribution, independently of the others.

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i)$$

i.i.d.

Coin flips. Patients picked at random.

Not i.i.d.

Prices day after day. Ten photos of the same patient.

Maximum likelihood

A model with parameters

A family of distributions. Learning picks one member.

$$p(x \mid \theta)$$

Coin

$\theta = P(\text{heads})$

Normal

$\theta = (\mu, \sigma)$

Linear regression

$\theta = (w_0, w_1, \ldots, w_p)$

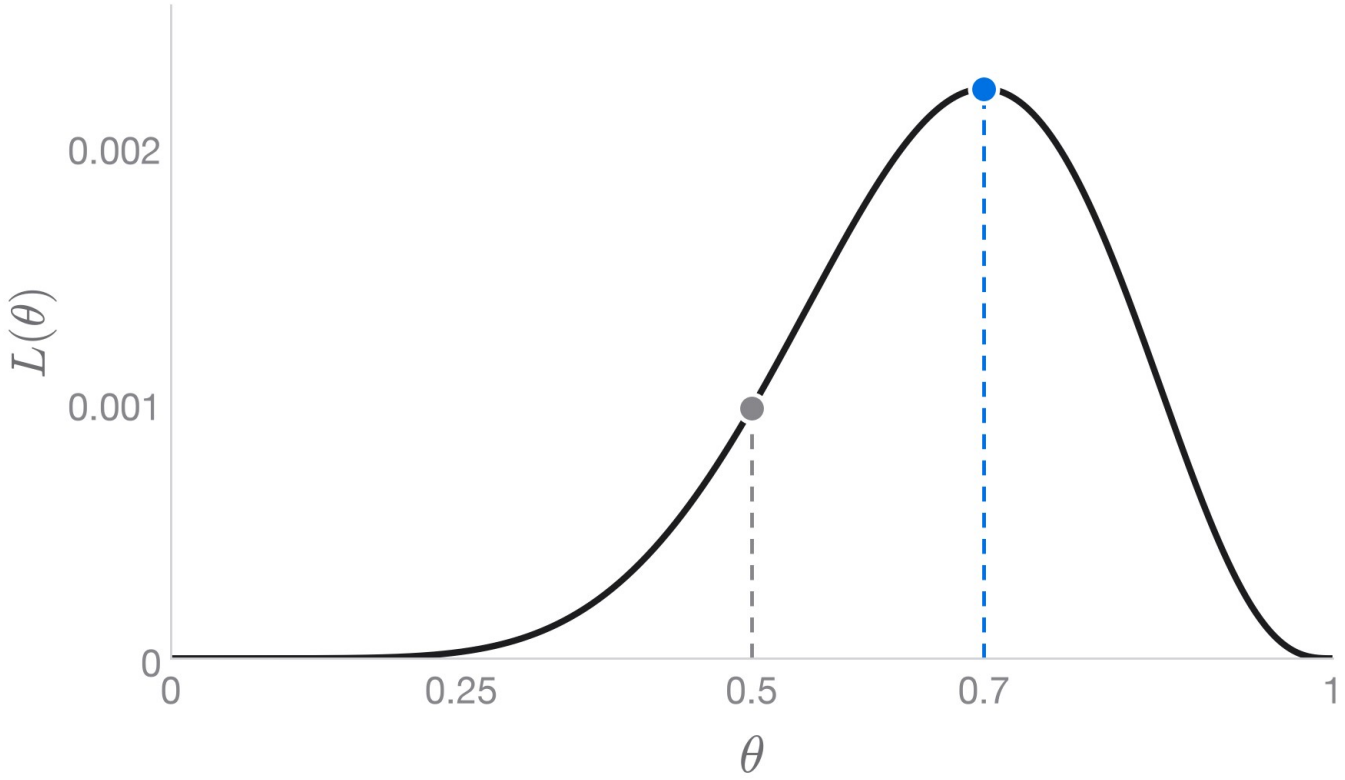
kNN keeps the whole training set instead: a non-parametric model.

10 flips, 7 heads.
How biased is the coin?

Most people answer 0.7. Maximum likelihood is the rule behind that answer.

Likelihood: how probable is this data?

Fix the data. Try a value of θ. Compute the probability of what we saw.



$$L(\theta) = \theta^7 (1-\theta)^3$$

$$L(0.5) = 0.00098$$

$$L(0.7) = 0.00222$$

The data is 2.3 times more probable with 0.7 than with a fair coin.

Probability and likelihood

Probability

θ is fixed, the data varies

Adds up to 1 over all possible data

Likelihood

The data is fixed, θ varies

Does not add up to 1 over θ

Independent flips: multiply

With i.i.d. data the probability of the whole dataset is a product.

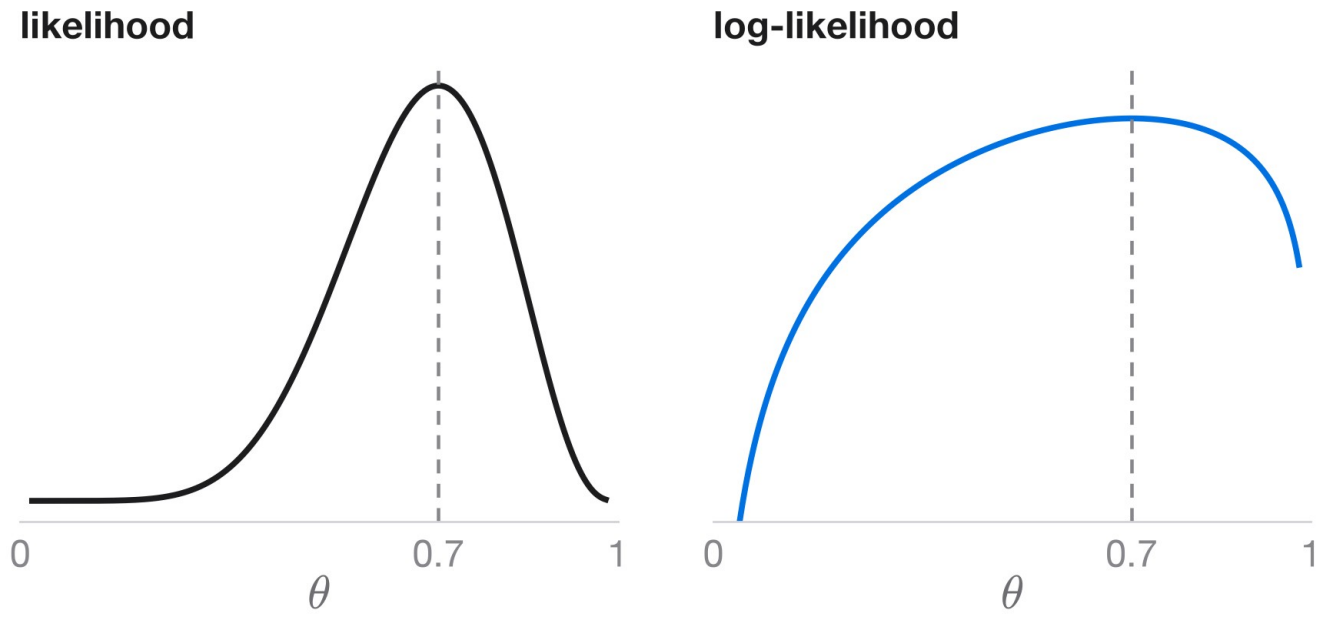
$$L(\theta) = p(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta)$$

H H T H H T H H T H

$$\theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot \theta \cdot (1-\theta) \cdot \theta = \theta^7 (1-\theta)^3$$

Products vanish. Take the logarithm

The logarithm is increasing, so the maximum stays in the same place.



1,000 flips

$$0.7^{700} \cdot 0.3^{300} \approx 10^{-265}$$

2,000 flips: zero in floating point

$$0.7^{1400} \cdot 0.3^{600} \approx 10^{-531}$$

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta)$$

Maximum likelihood estimation

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \log p(x_i \mid \theta)$$

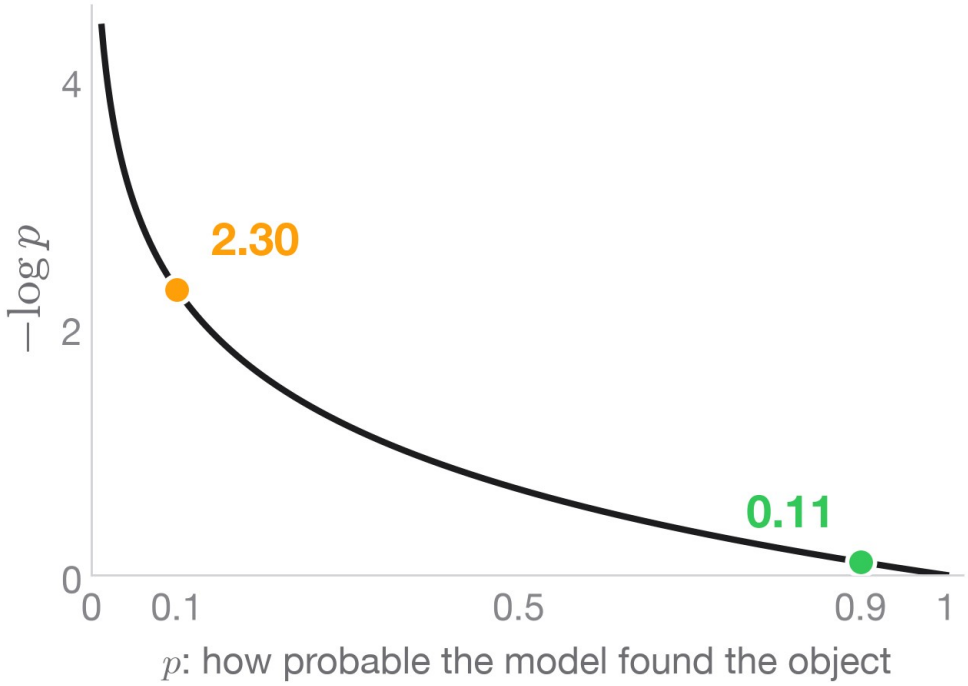
$p(x_i \mid \theta)$	how probable object *i* is under the model
$\hat{\theta}$	the value that makes the observed data most probable
$\log$	the natural logarithm; any base gives the same answer

The same thing as a loss

Optimisers minimise, so flip the sign: the negative log-likelihood.

$$\mathrm{NLL}(\theta) = -\frac{1}{n}\sum_{i=1}^{n} \log p(x_i \mid \theta)$$

An object the model found likely costs little. A surprising one costs a lot.



The coin, by calculus

Write the log-likelihood. Set its derivative to zero.

$$\ell(\theta) = k \log \theta + (n - k) \log(1 - \theta)$$

$$\ell'(\theta) = \frac{k}{\theta} - \frac{n - k}{1 - \theta} = 0$$

$$\hat{\theta} = \frac{k}{n} = \frac{7}{10}$$

0.7

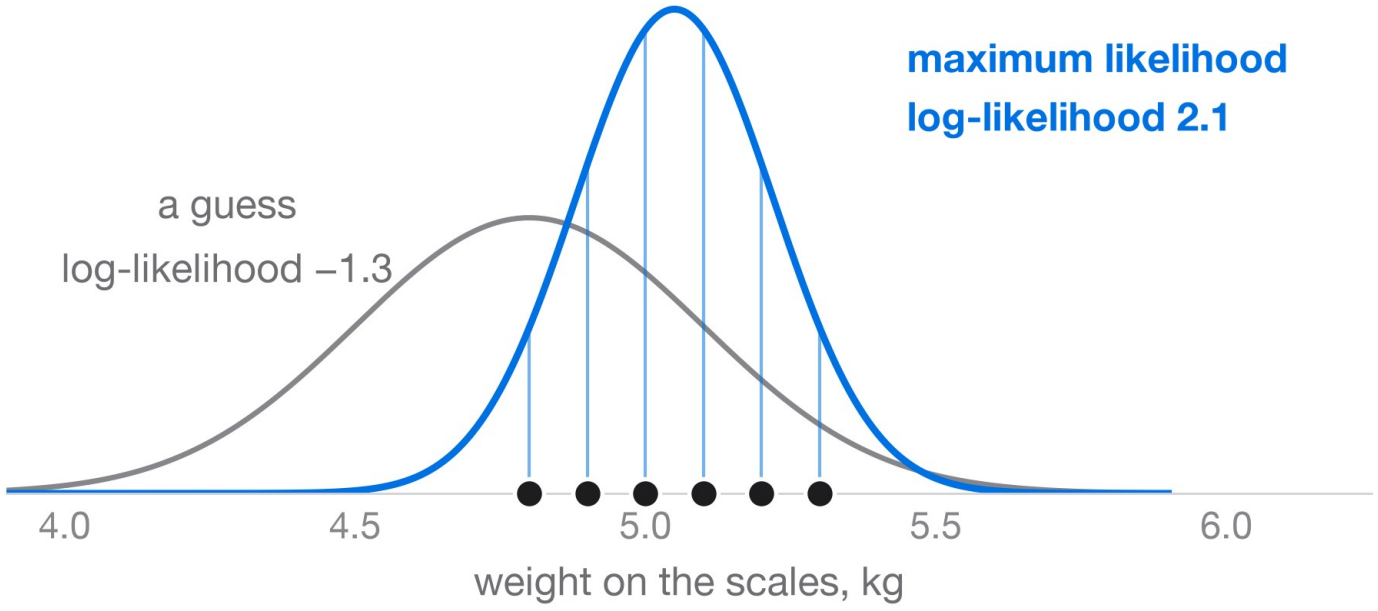
The fraction of heads, as everyone guessed.

A melon on market scales

Six weighings of one melon. Each is the true weight plus noise.

4.9 5.2 5.0 5.3 4.8 5.1 kg

$$x_i = \mu + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$



Which bell fits best?

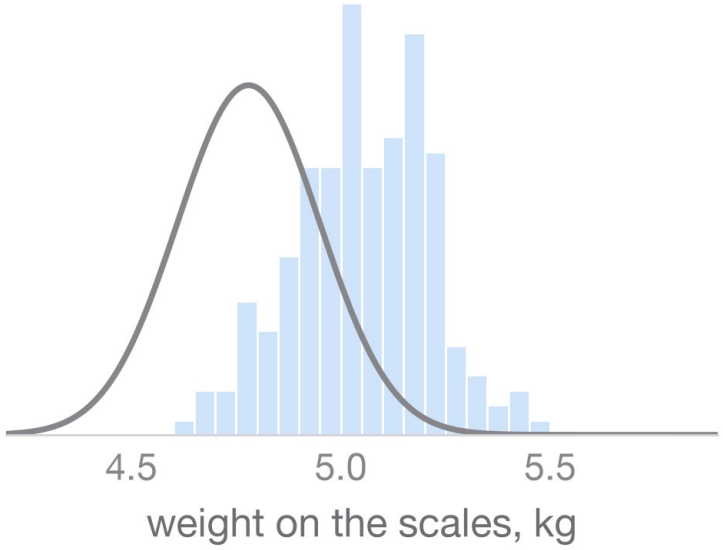
Move μ and σ until the six points are as probable as they can be.

The best bell lies on the data

Three bells over 200 weighings. The likelihood peaks when the bell covers the data.

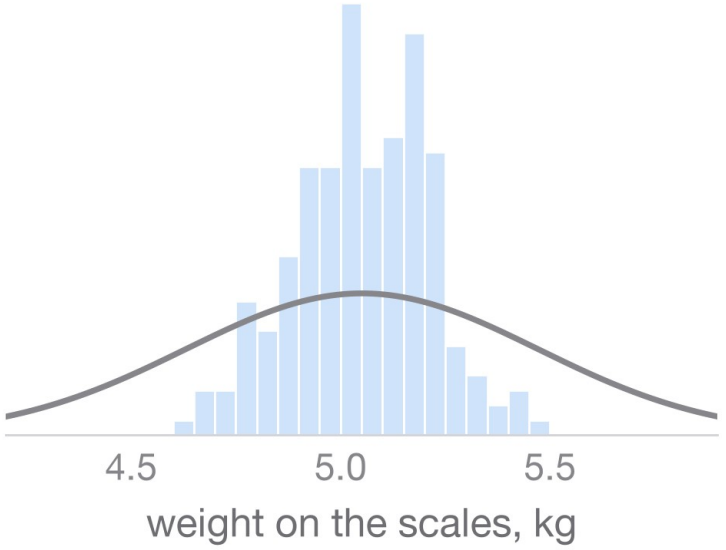
Too far left

log-likelihood –173



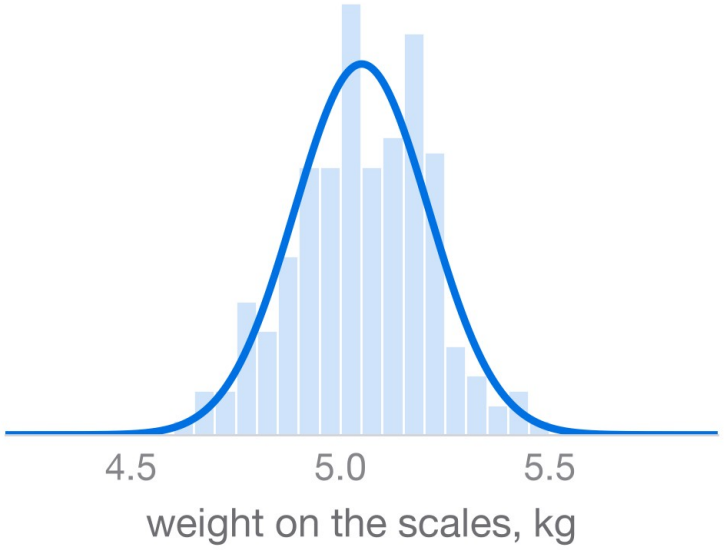
Too wide

log-likelihood –25



Maximum likelihood

log-likelihood 82



The normal distribution, by calculus

$$\ell(\mu, \sigma) = -n \log\left(\sigma\sqrt{2\pi}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i = 5.05$$

the sample mean

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\mu})^2, \qquad \hat{\sigma} = 0.17$$

the mean squared deviation

To find the best *μ* we minimised a sum of squares.

Maximum likelihood can be biased

It divides by *n*. On average that comes out a little too small.

0.171

maximum likelihood

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

0.187

unbiased estimate

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

With more data the gap disappears.

The recipe

1. Choose a family of distributions with parameters
2. Write the log-likelihood of the data
3. Maximise it, by calculus or with an optimiser

Why trust maximum likelihood

Consistent

With more data the estimate approaches the true value.

Efficient

For large samples no unbiased estimator varies less.

Invariant

The estimate of a function of the parameter is that function of the estimate.

Most models are trained this way

Choose a distribution for the target. Its negative log-likelihood is the loss.

the target	its distribution	the loss
A number	normal	**squared error**
A class	categorical	**cross-entropy**
The next word	categorical over the vocabulary	**cross-entropy**

Other ways to estimate

MAP

Add a prior belief about the parameters to the likelihood.

Bayesian

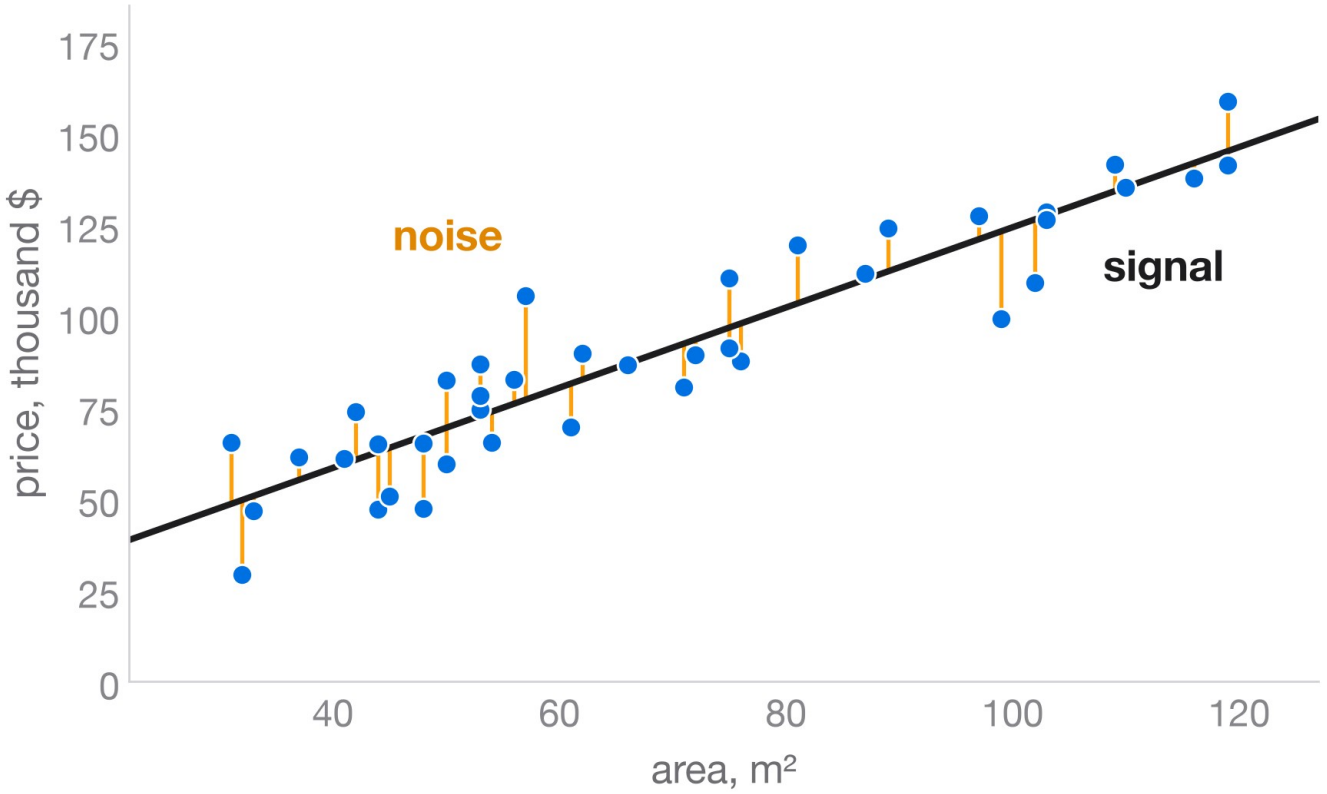
Keep a whole distribution over the parameters instead of one value.

Moments

Match the model's mean and variance to the sample's.

Why squared error

A target is signal plus noise

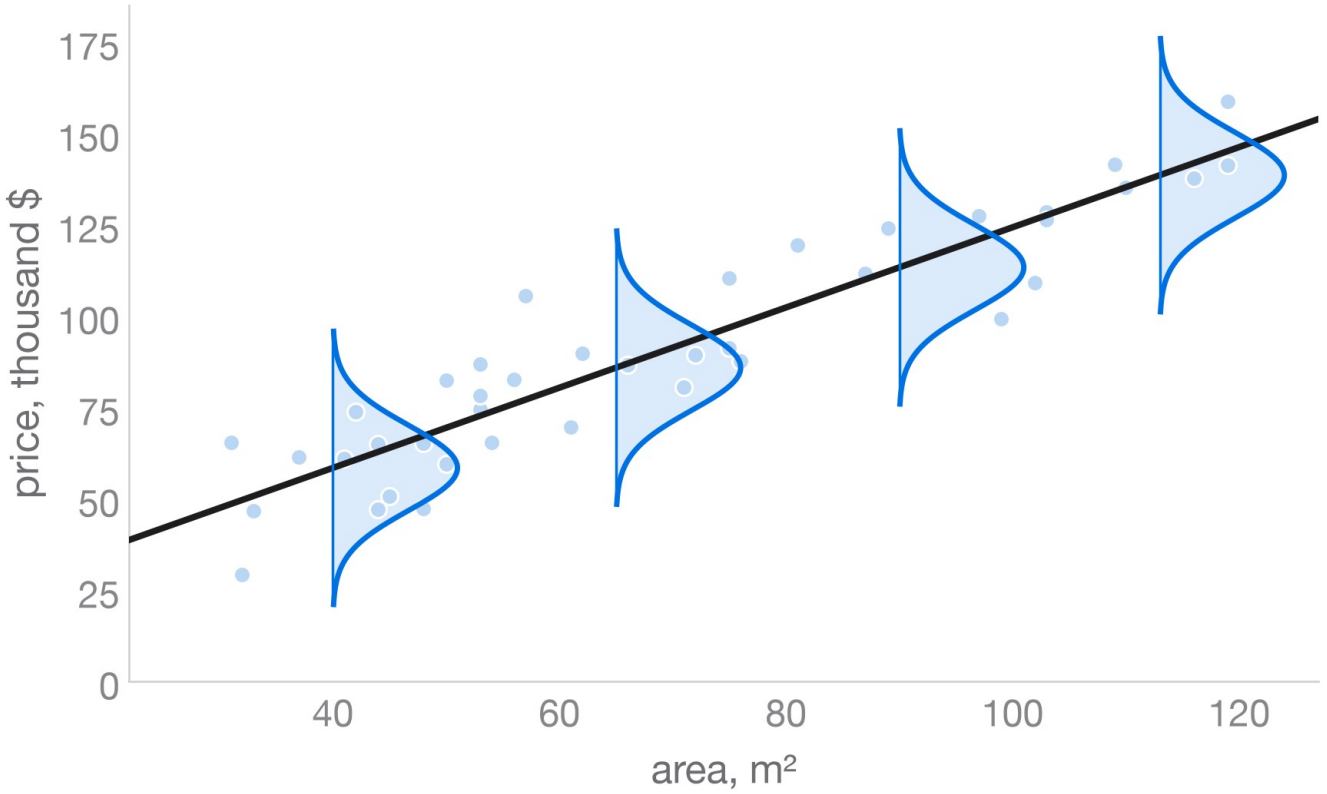


$$y = f(x) + \varepsilon$$

Signal: what the features explain.

Noise: everything else. Zero on average.

Assume the noise is normal



$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2), \text{ i.i.d.}$$

then

$$y_i \mid x_i \sim \mathcal{N}\big(f_w(x_i), \sigma^2\big)$$

For every area, the price is a bell centred on the line.

The log-likelihood of the data

One flat first. Then the logarithm, and the sum over all flats.

$$p(y_i \mid x_i, w) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y_i - f_w(x_i))^2}{2\sigma^2}\right)$$

$$\ell(w) = -n \log\left(\sigma\sqrt{2\pi}\right) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - f_w(x_i)\right)^2$$

do not depend on the weights

the sum of squared errors

Most likely means least squares

$$\arg\max_w \ \ell(w) \ = \ \arg\min_w \ \sum_{i=1}^{n} \big(y_i - f_w(x_i)\big)^2$$

The constant and the factor do not move the maximum.

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} \big(y_i - f_{\hat{w}}(x_i)\big)^2 = \mathrm{MSE}$$

the best estimate of the noise level

Normal noise plus maximum likelihood gives least squares

The loss follows from what we assume about the noise.

Other noise, other loss

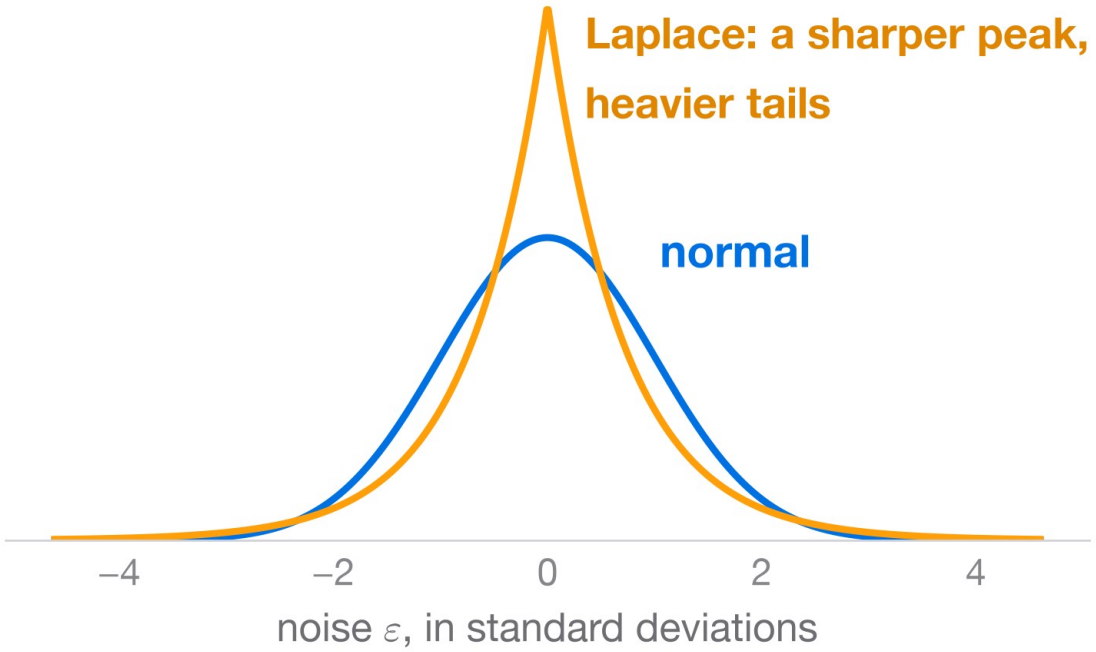
Maximum likelihood turns each noise model into its own loss.

Normal noise

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$ squared error

Laplace noise

$$\sum_{i=1}^{n}|y_i - \hat{y}_i|$$ absolute error



One outlier pulls the line



Squared error follows the mean.

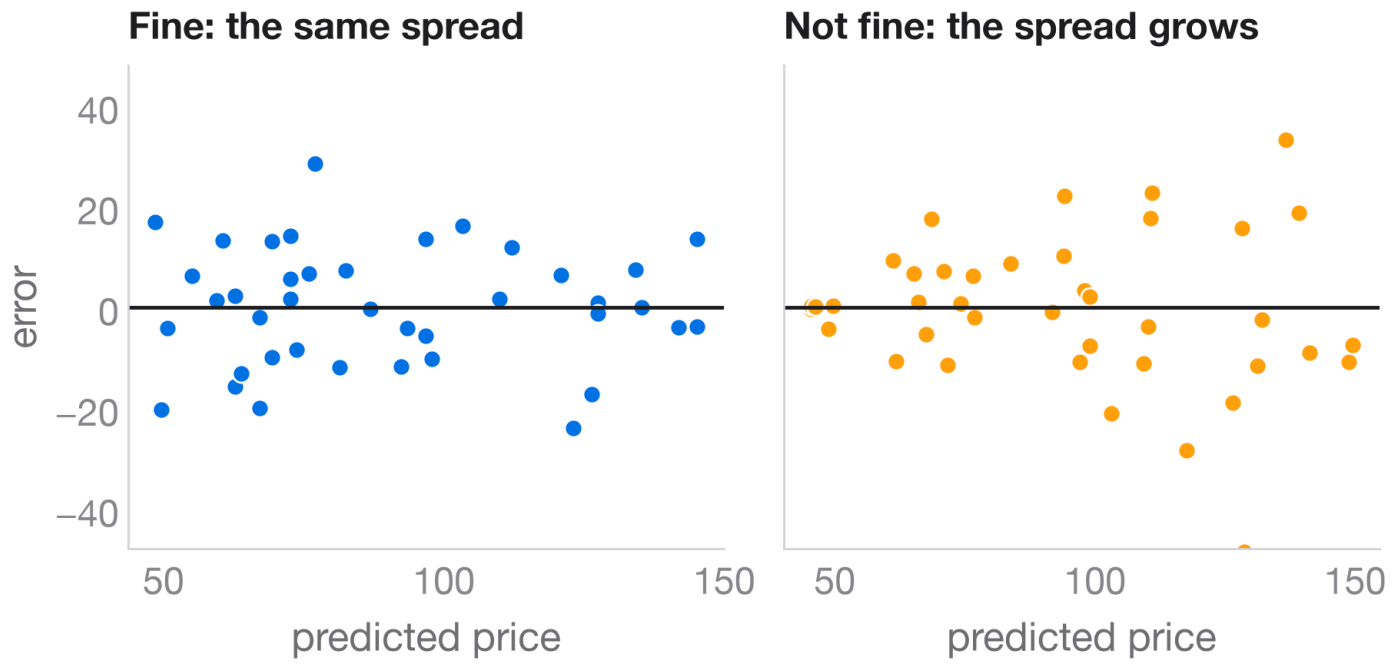
A miss ten times bigger costs a hundred times more.

Absolute error follows the median.

A miss ten times bigger costs ten times more.

Check the residuals

Plot the errors against the predictions. The noise should look the same everywhere.



Centred on zero

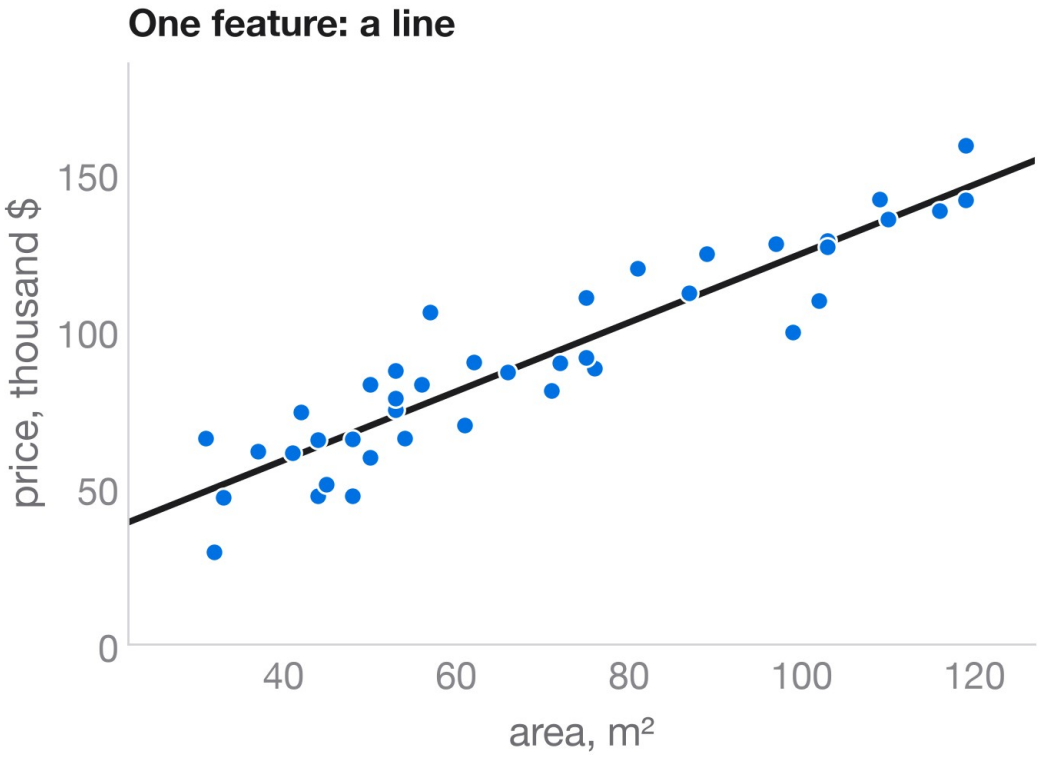
The same spread everywhere

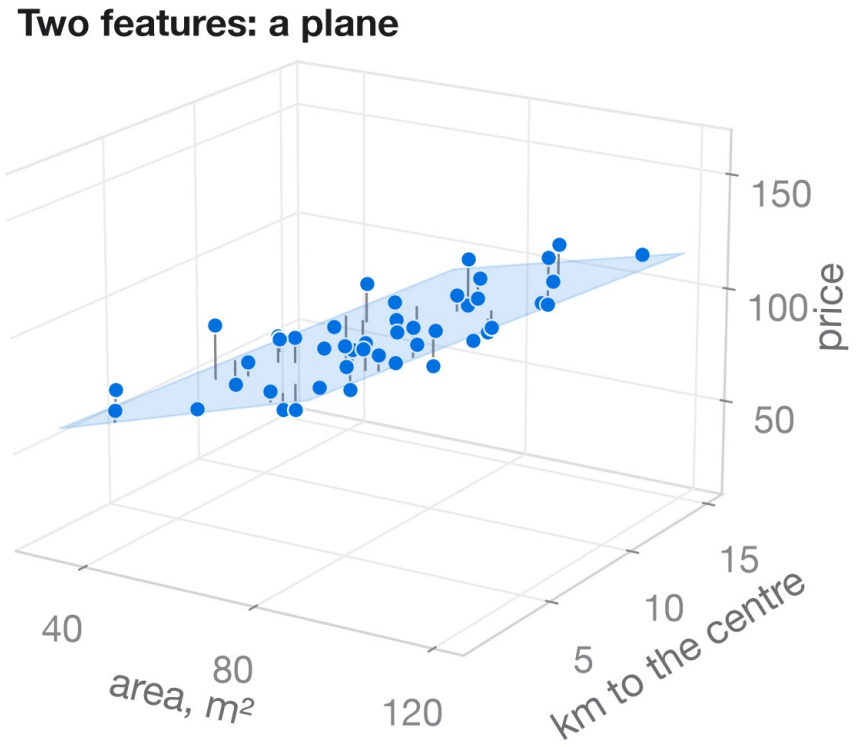
No pattern from one object to the next

Linear models

A linear model is a weighted sum

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_p x_p$$





A column of ones holds the bias

Add a feature that always equals 1. Its weight is the bias.

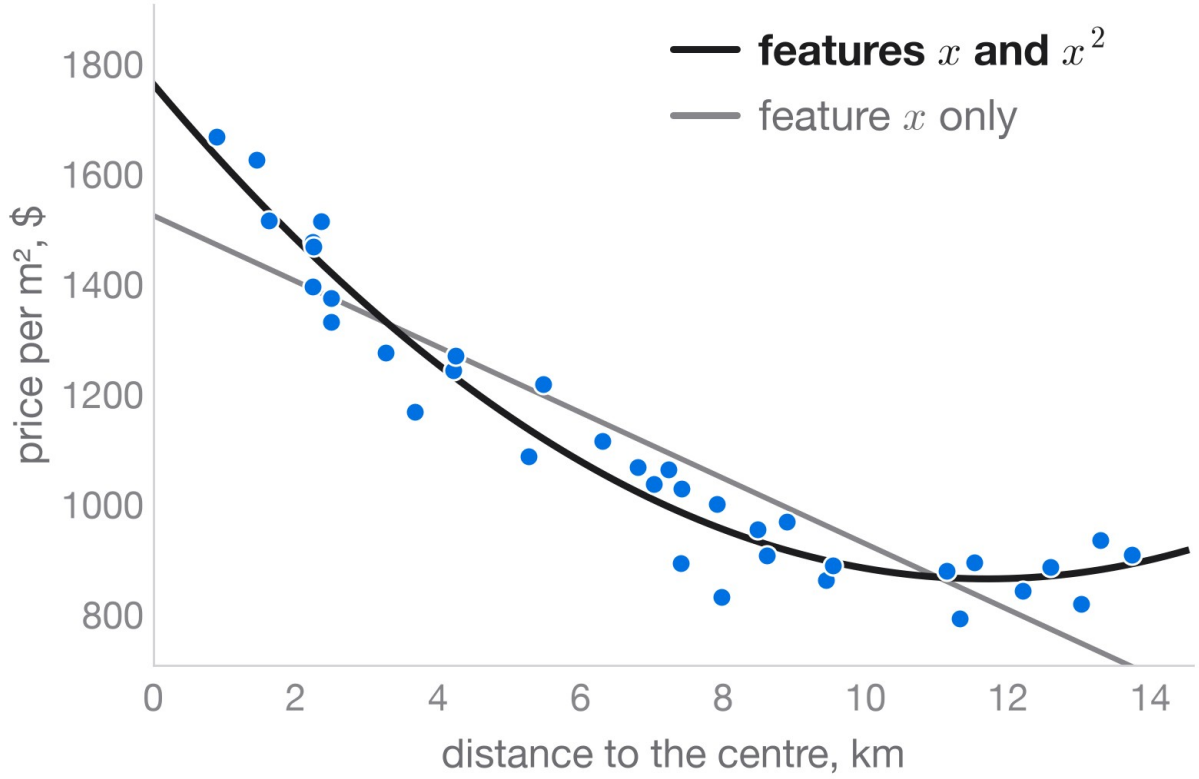
ones	area, m²	price
1	89	124.1
1	119	158.7
1	50	82.5
1	32	29.3

$$\hat{y} = w^\top x, \qquad x = (1, x_1, \ldots, x_p)$$

$$\hat{y} = Xw$$ all the predictions at once

Linear in the weights

Features can be squares, logarithms, products. The model stays linear in *w*.



$$\hat{y} = w_0 + w_1 x + w_2 x^2$$

A curve on the plot. Still a weighted sum of two features.

Linear models are everywhere

Regression

Predict a number. Today.

Classification

Logistic regression, later in the course.

Fewer dimensions

PCA, later in the course.

Neural networks

Every layer starts with a weighted sum.

Why start with linear models

Fast

Trained in milliseconds, even on millions of rows.

Readable

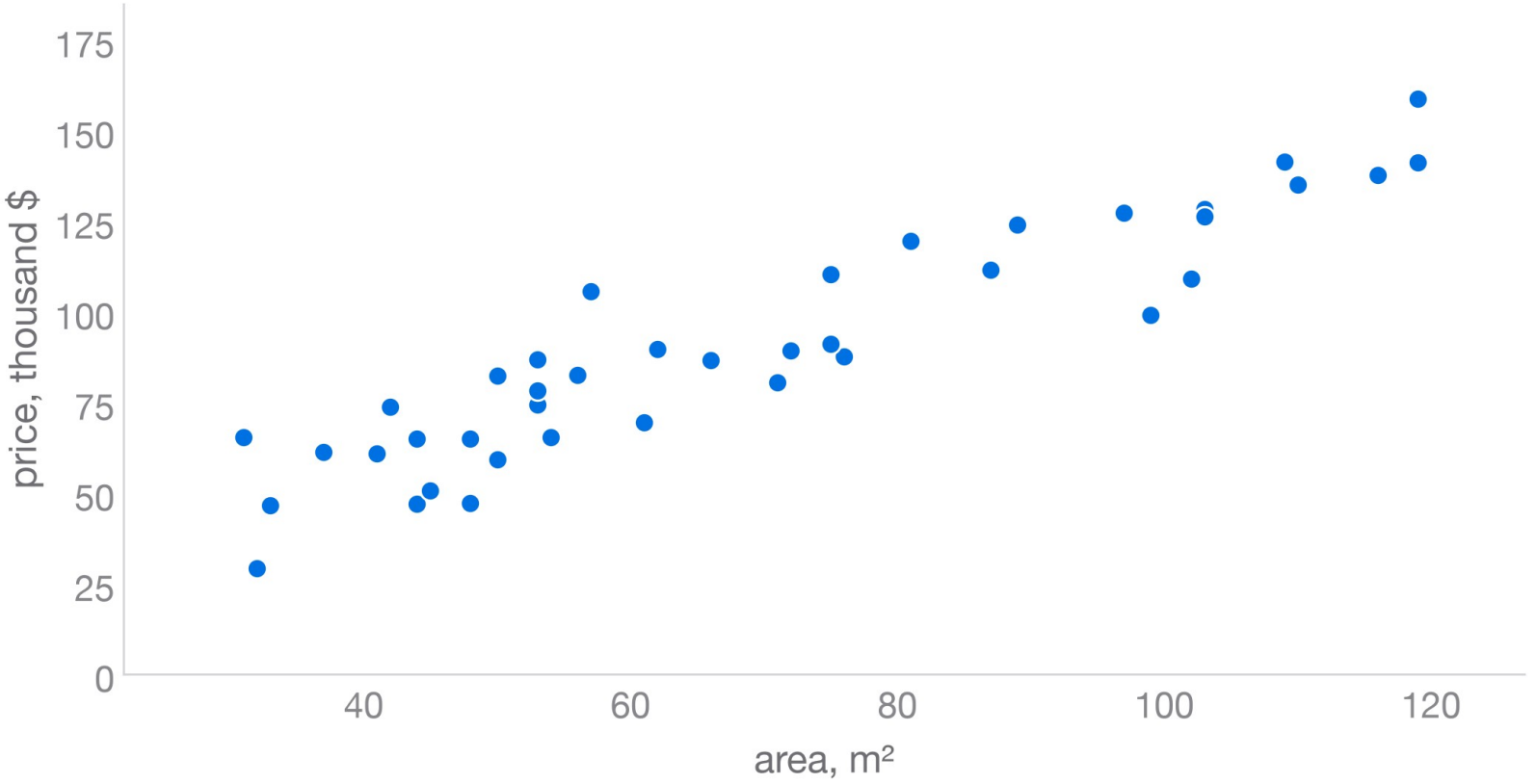
Each weight says how a feature moves the prediction.

A baseline

Strong on tables. Beat it before you try anything bigger.

Linear regression

Flats in Tashkent: price from area

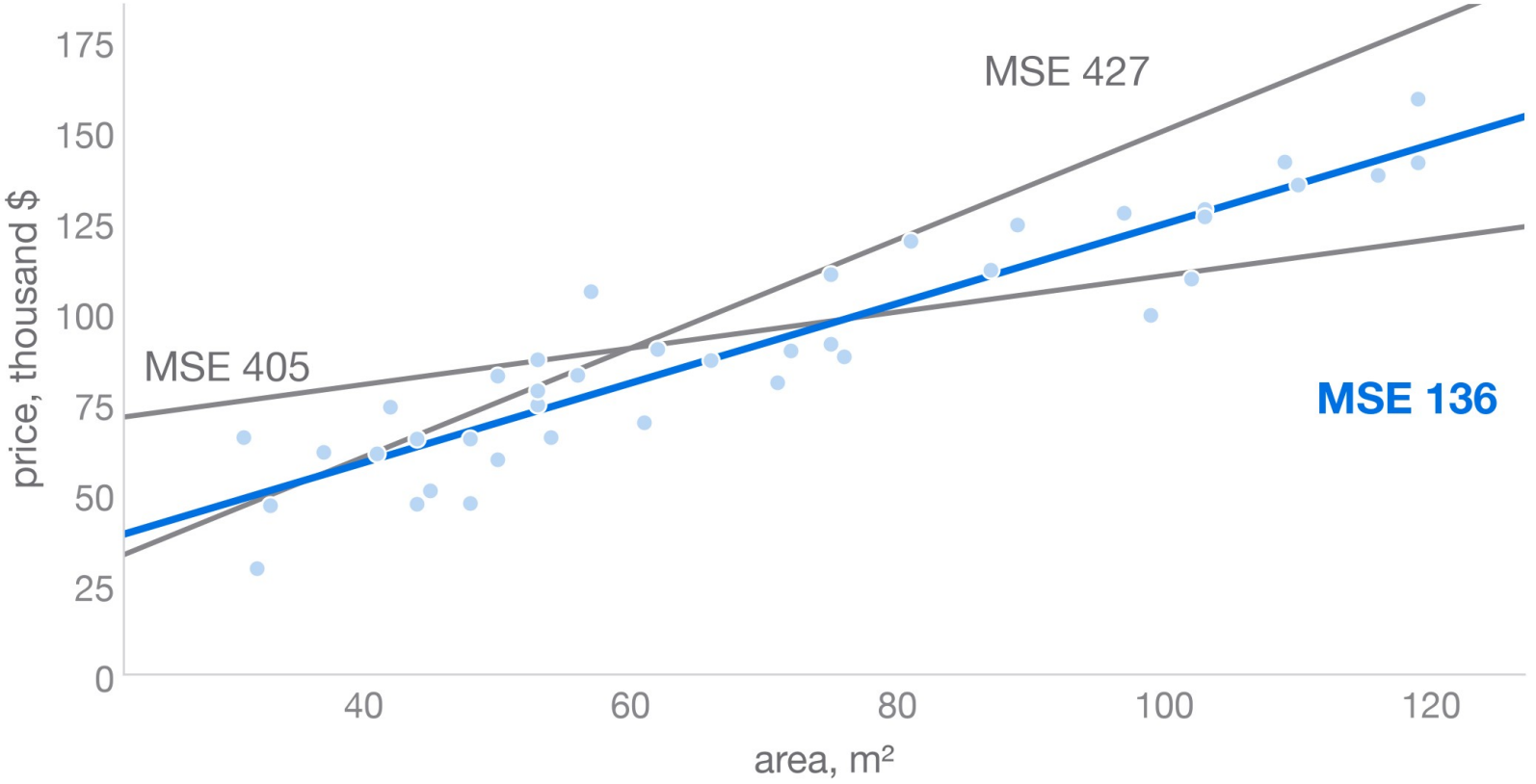


40

flats

Area in m², price in thousand dollars.

Every line has a loss

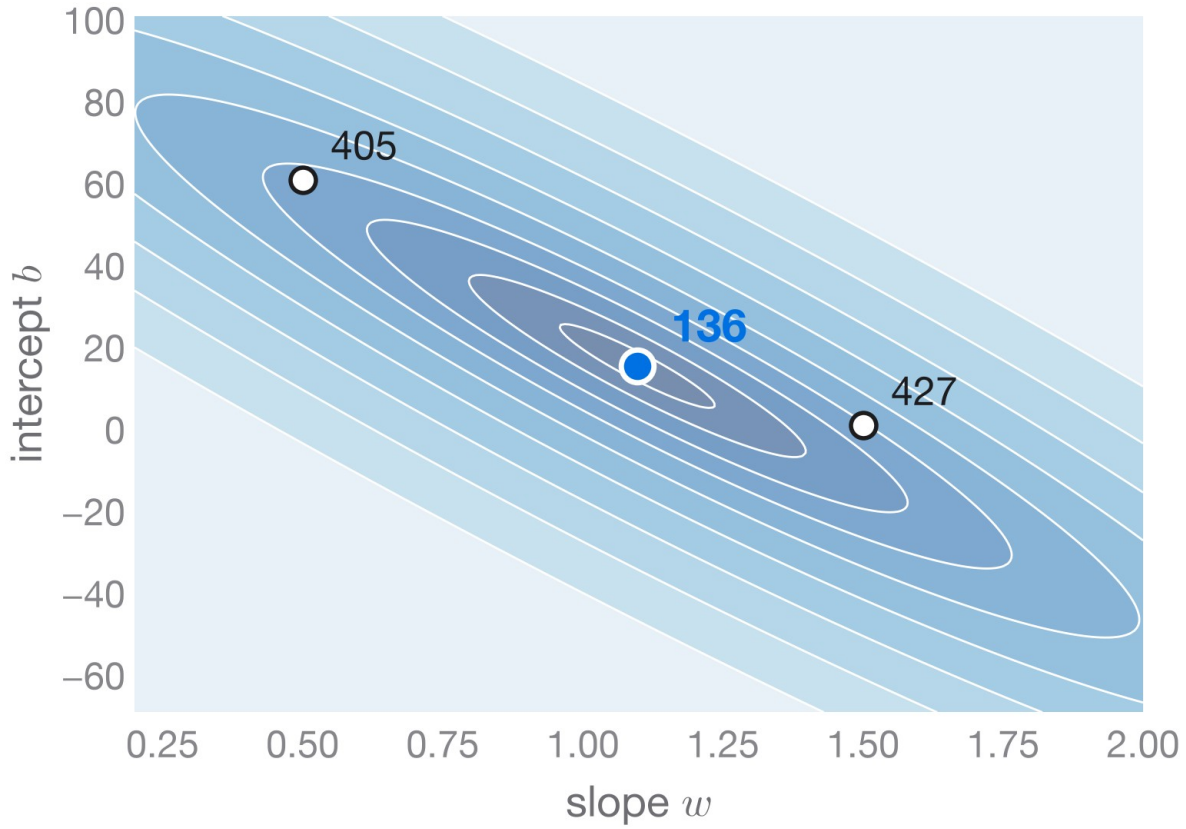


$$\hat{y} = w \cdot \text{area} + b$$

One line is one pair of weights.

The best line has the lowest MSE.

The loss is a bowl



$$L(w,b) = \frac{1}{n}\sum_{i=1}^{n}(y_i - wx_i - b)^2$$

Each point of the map is a line.

A convex bowl: one minimum, no traps.

The same loss with matrices

$$L(w) = \frac{1}{n} \|y - Xw\|^2$$

$X \in \mathbb{R}^{n \times (p+1)}$	the features, with the column of ones
$y \in \mathbb{R}^n$	the targets
$w \in \mathbb{R}^{p+1}$	the weights, bias included

Set the gradient to zero

The bottom of the bowl is where the gradient vanishes.

$$\nabla L(w) = -\frac{2}{n} X^\top (y - Xw) = 0$$

$$X^\top X\, w = X^\top y$$

the normal equations

$$\hat{w} = (X^\top X)^{-1} X^\top y$$

the least squares solution

The five students again

The weights from last week come out of the formula.

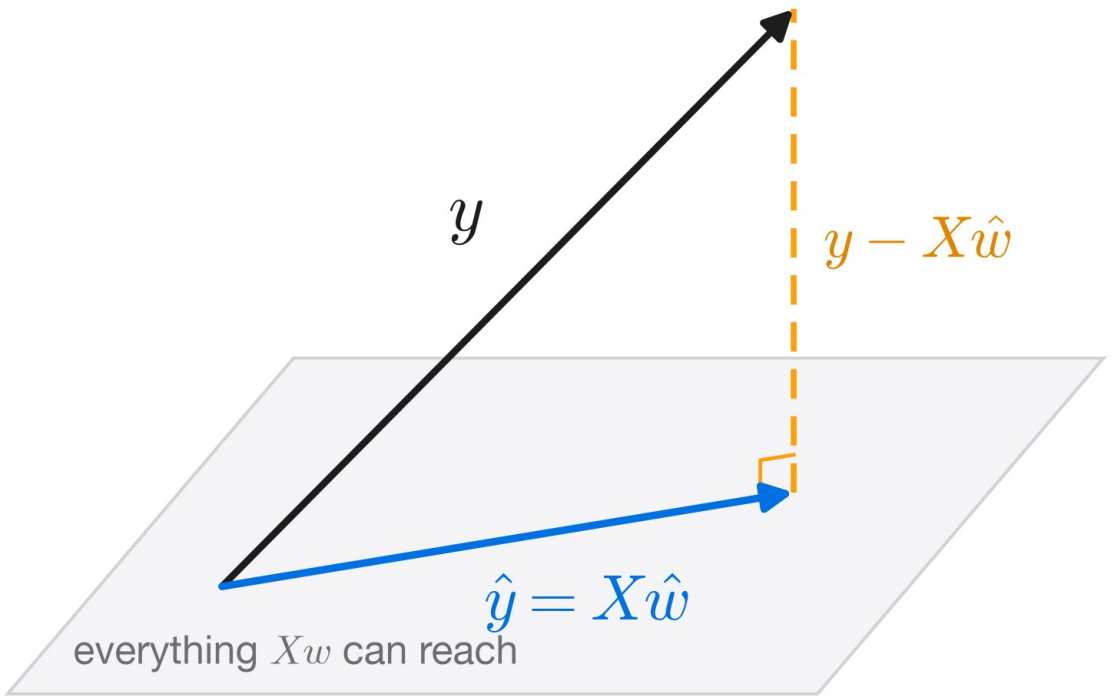
name	statistics	python	ML mark
Aziz	5	4	5
Malika	4	5	4
Timur	5	5	5
Dilnoza	3	4	5
Sardor	3	3	2

$$X^\top X = \begin{pmatrix} 84 & 86 \\ 86 & 91 \end{pmatrix}, \qquad X^\top y = \begin{pmatrix} 87 \\ 91 \end{pmatrix}$$

$$\hat{w} = (X^\top X)^{-1} X^\top y = \begin{pmatrix} 0.37 \\ 0.65 \end{pmatrix}$$

Two features and no bias, as last week.

Geometry: a projection



A weighted sum of the columns can only reach a plane.

The closest point to the targets is their projection.

$$X^\top (y - X\hat{w}) = 0$$

The error is perpendicular to every feature.

BLUE

Least squares is the best linear unbiased estimator

If the noise has zero mean, equal variance and no correlation, no other linear unbiased estimator varies less. Normality is not needed for this.

Where the formula hurts

Cost

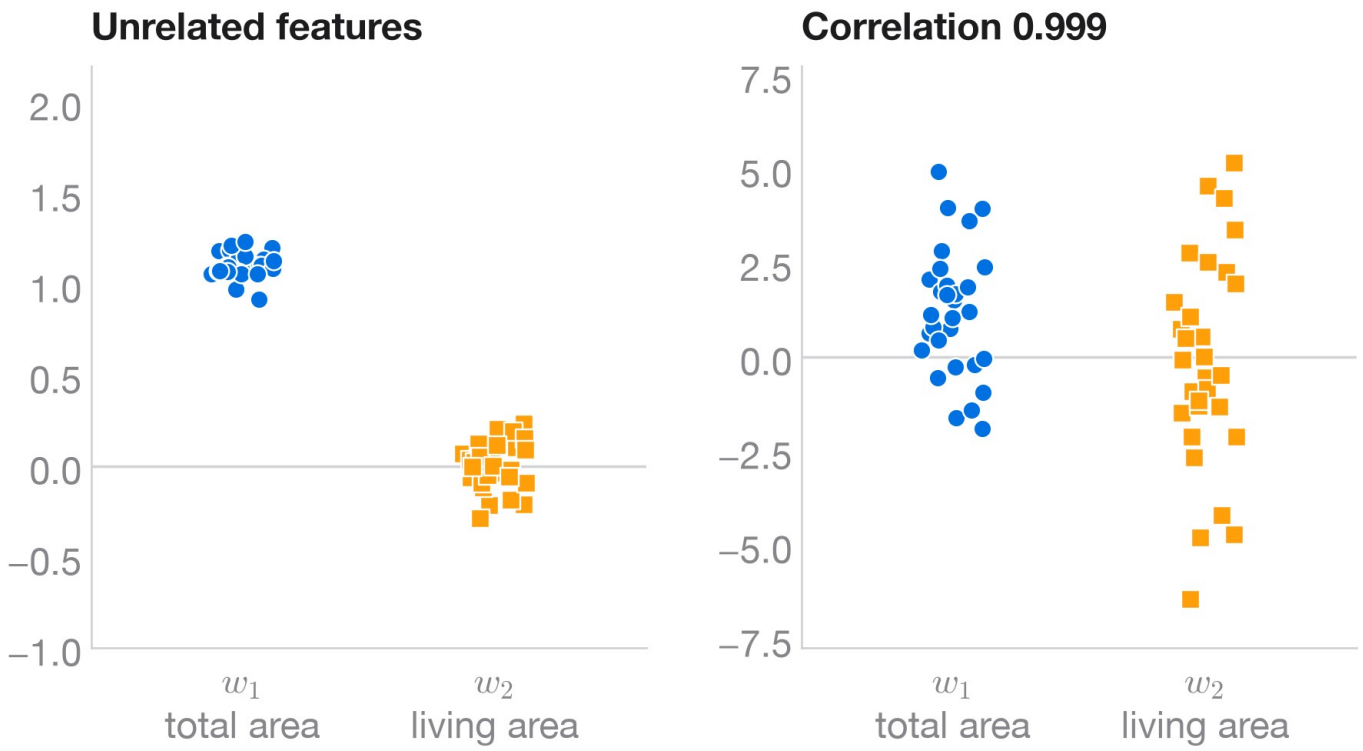
$$O(np^2 + p^3)$$

Nothing for a hundred features.
Hopeless for a million.

Stability

When features repeat each other, the matrix to invert is almost singular and the weights blow up.

Features that repeat each other

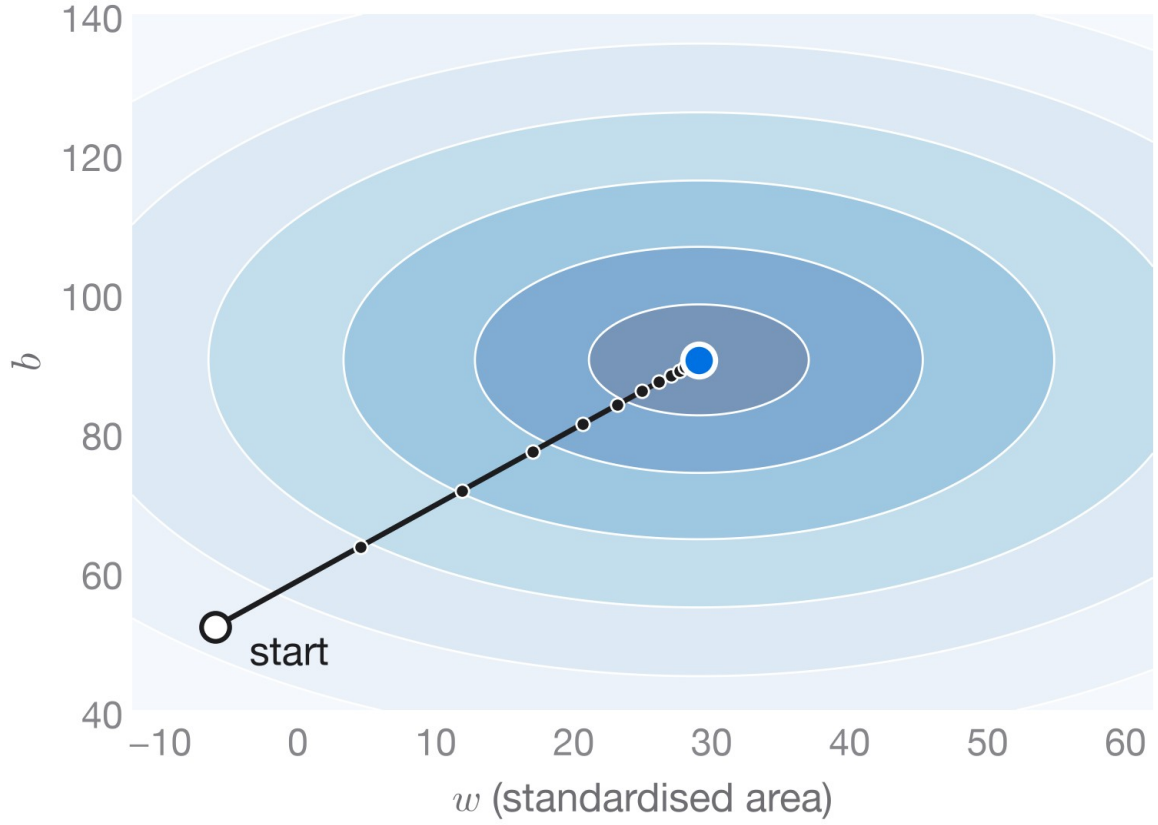


Total area and living area tell the same story.

Thirty samples of 40 flats. The fit cannot decide how to share the weight between the twins.

The predictions stay fine. The weights do not.

Gradient descent: walk downhill



$$w \leftarrow w - \eta \nabla L(w)$$

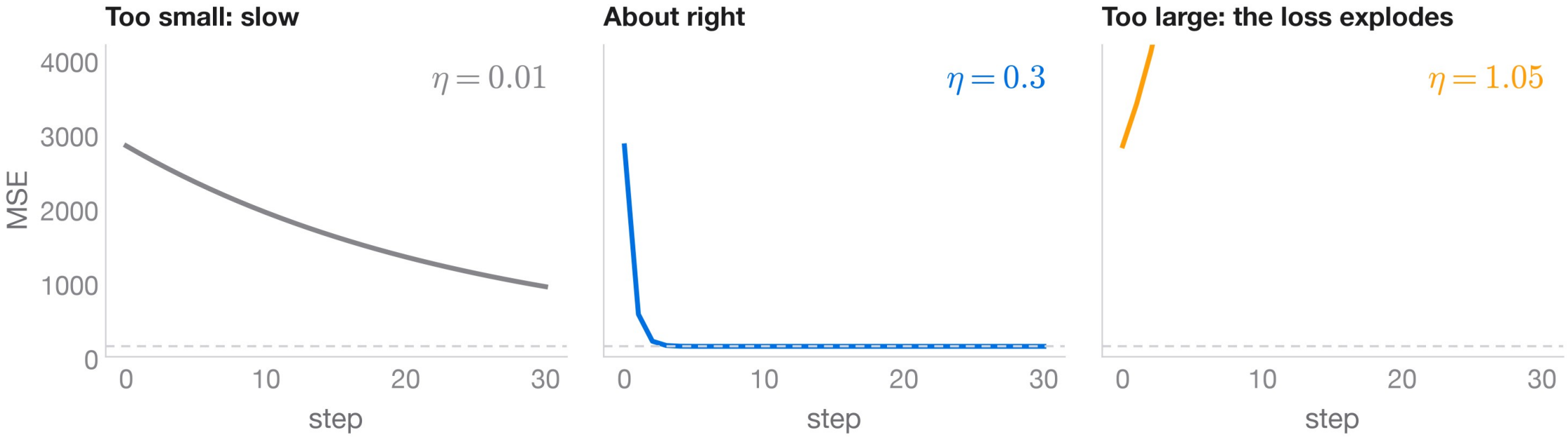
Start anywhere. Step against the gradient. Repeat.

η is the learning rate, the size of a step.

$$\nabla L(w) = -\frac{2}{n} X^\top (y - Xw)$$

The learning rate decides

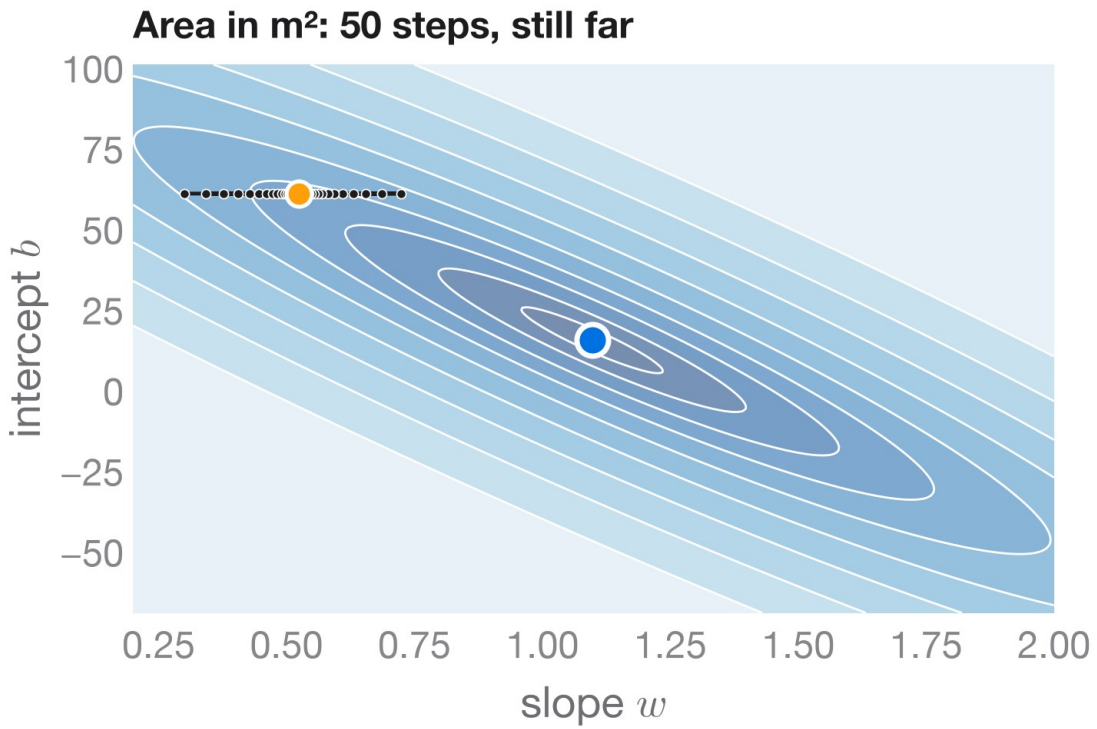
Thirty steps from the same start, with three learning rates.

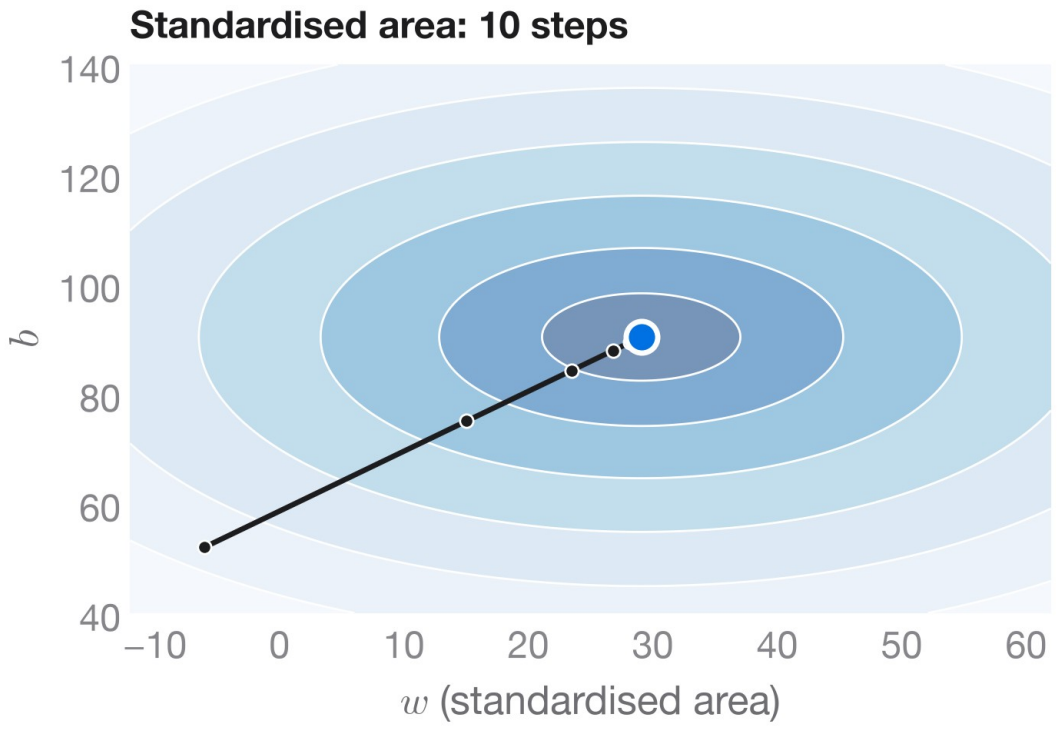


Try a few values on a logarithmic grid: 0.001, 0.01, 0.1, 1.

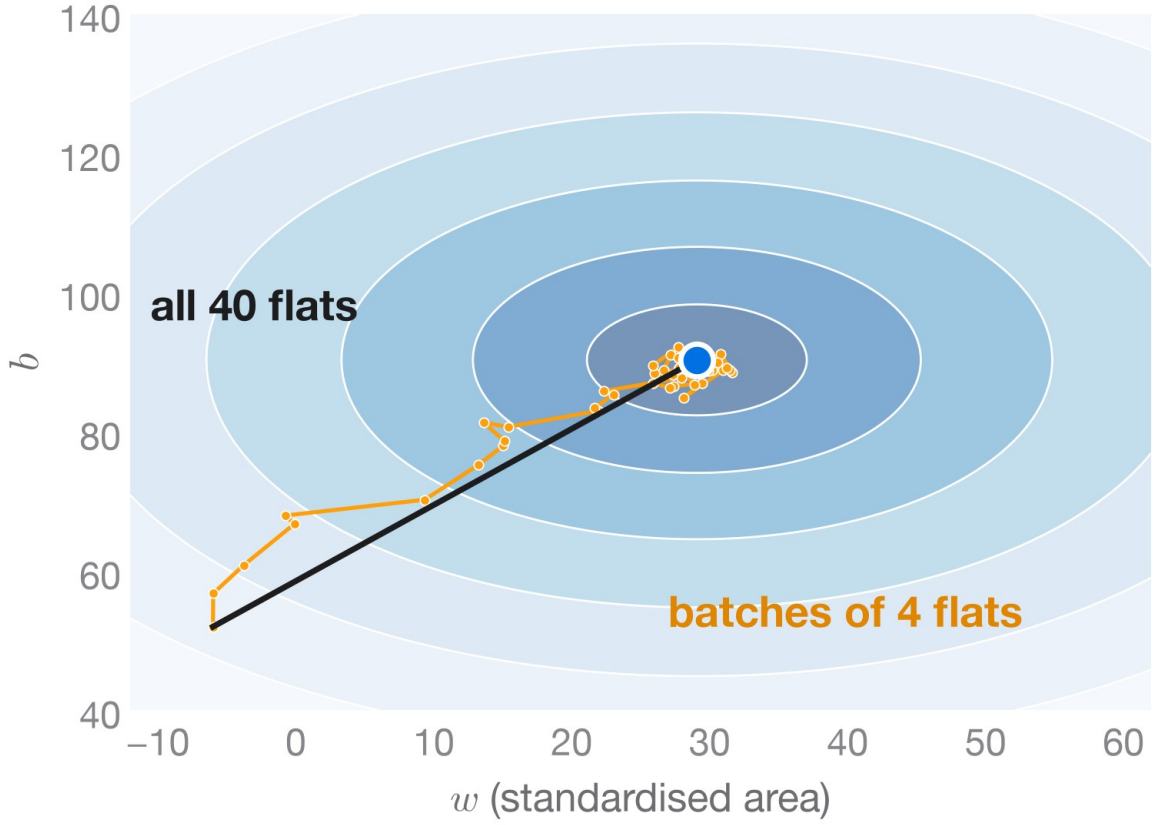
Scale the features first

As with kNN: raw units stretch the bowl into a ravine.





Big data: one small batch at a time



$$w \leftarrow w - \eta \nabla L_B(w)$$

Estimate the gradient on a small random batch.

Each step is cheap and a little noisy.

This is stochastic gradient descent.

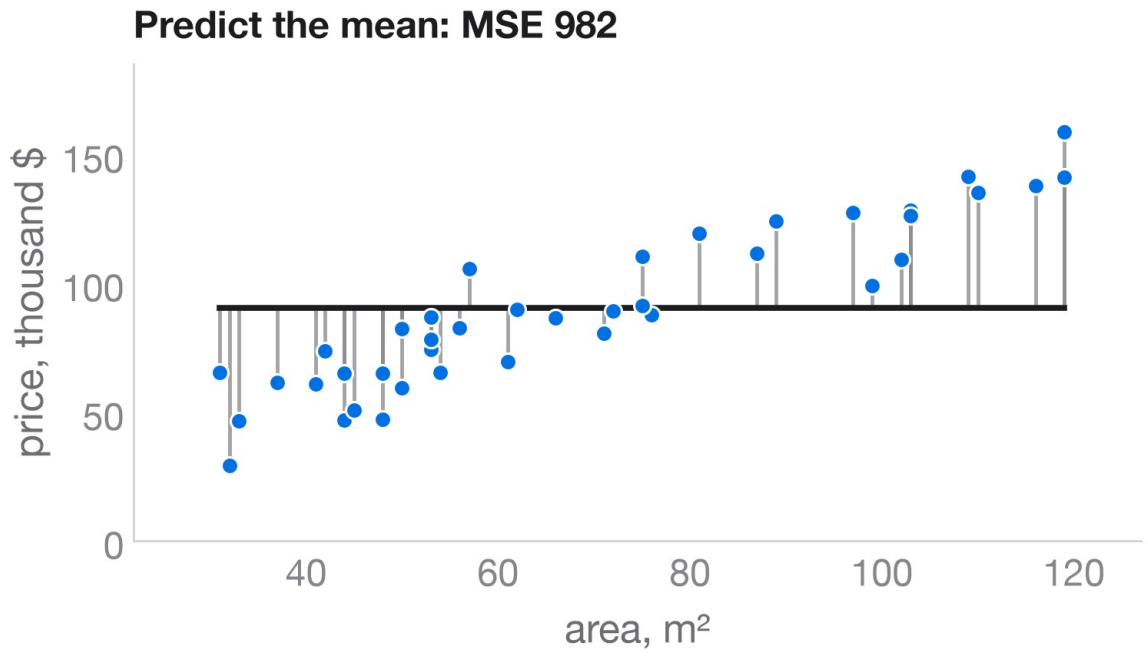
Metrics for regression

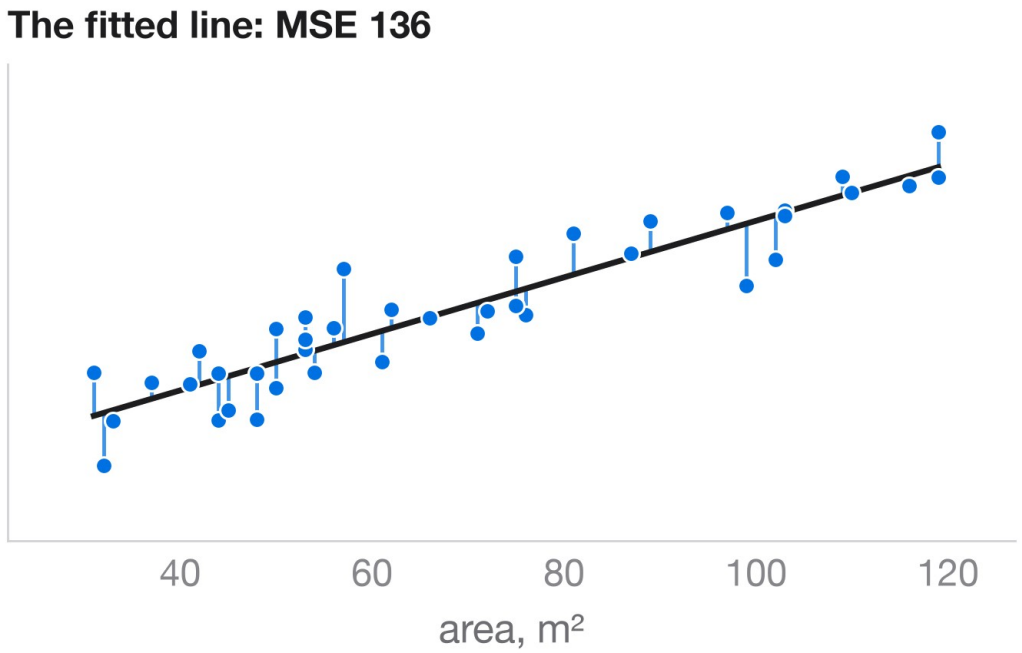
We train on a loss and report metrics.

metric	formula	what it says	our flats
MSE	$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	the loss itself, in squared units	**136.5**
RMSE	$\sqrt{\text{MSE}}$	a typical miss, in the units of the target	**11.7**
MAE	$\frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i\rvert$	the average miss, calmer about outliers	**9.4**
R squared	$1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$	1 is perfect, 0 is the mean baseline	**0.86**

R squared: compare with the mean

Baseline first: the simplest model predicts the mean price for every flat.





$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} = 1 - \frac{136.5}{981.8} = 0.86$$

Reading the weights

$$\widehat{\text{price}} = 14.5 + 1.10 \cdot \text{area}$$

Each extra square metre adds about 1,100 dollars.

14.5 is the prediction at zero area. No flat is that small, so do not read it.

To compare the weights of different features, standardise the features first.

Linear regression in scikit-learn

```
from sklearn.linear_model import LinearRegression

model = LinearRegression().fit(X_train, y_train)

model.intercept_, model.coef_     # 14.5, [1.10]
model.predict(X_new)              # prices of new flats
model.score(X_val, y_val)         # R squared on validation data
```

Revise

Likelihood: the probability of the data as a function of the parameters.

Maximum likelihood: maximise the log-likelihood, or minimise NLL.

With normal noise, the maximum likelihood fit is least squares.

Linear regression: normal equations for small data, gradient descent for big.

Exit quiz

Scan the code.

Enter your full name. One question at a time, 30 seconds each.

If time allows, we then play the multiplayer game Neighbours.



165.22.93.84

Practice

Seminar

One notebook: linear regression, from likelihood to scikit-learn.

Interactive

Six pages at 165.22.93.84/viz: likelihood, noise, the loss bowl, gradient descent.

Self-check

Practice questions with explanations, on the same site.

Links to the notebook are in the course chat.

To read

Books

Bishop, Pattern Recognition and Machine Learning: 1.2 and 3.1

Hastie, Tibshirani, Friedman, The Elements of Statistical Learning: chapter 3

Deisenroth, Faisal, Ong, Mathematics for Machine Learning: chapters 8 and 9

Online

StatQuest on YouTube: Maximum Likelihood, clearly explained

statlect.com: maximum likelihood estimation

Yandex ML handbook, in Russian: linear models

Next time: linear models, part two

Bring your questions about the seminar notebook.

Questions?

Makhmudjon Sodikov, YMLS Uzbekistan

YMLS



girafe ai